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1 Background

1.1 Education

PhD in Theoretical Physics 2016-2021, The University of Alabama - USA, Tuscaloosa

Study Abroad in JYPE 2014-2015, Tohoku University - Japan, Sendai

BS in Engineering Physics 2012-2016, The Colorado School of Mines - USA, Golden

1.2 Appointments/Employments

Assistant Professor April 2026 - Present

National Institute of Technology Oyama College (Department of Electrical and Electronic Engineering)

Assistant Professor April 2025 - April 2026

National Institute of Technology Oyama College (Department of General Education)

JSPS Postdoc August 2023 - August 2024

Yamagata University

Technical Consultant January 2023 - May 2023

Quantum Data Center

Technical Consultant August 2022 - May 2023

quantumz.io

Postdoc January 2022 - June 2023

Henan University

Contract Tutor 2017 - 2021

Appliedtutoring.com

PhD Student 2016 - 2021

The University of Alabama

Graduate Teaching Assistant 2016 - 2021

The University of Alabama

2 Research

2.1 Publications (Including Preprints)

2026 - Revisiting the Wess-Zumino-Witten Term in Nuclear and Quark Matter under Magnetic Fields and Rotation

arXiv:[2607.04929](#)

2026 - Homogeneous anisotropic black branes with bianchi VI_h symmetry

[10.1140/epjc/s10052-026-15741-w](#) - arXiv:[2603.07993](#)

2026 - **Holographic QCD Matter: Chiral Soliton Lattices in Strong Magnetic Field**
[10.1007/JHEP02\(2026\)240](#) - [arXiv:2507.16897](#)

2024 - **Holographic Global Vortices with Novel Boundary Conditions**
[arXiv:2404.03212](#)

2024 - **Relativistic hydrodynamics under rotation: Prospects and limitations from a holographic perspective**
[10.1016/j.pnpnp.2024.104135](#) - [arXiv:2308.11686](#)

2023 - **Modes of the Sakai-Sugimoto soliton**
[10.1088/1751-8121/ad742c](#) - [arXiv:2306.00677](#)

2023 - **Convergence of hydrodynamics in rapidly spinning strongly coupled plasma**
[10.1103/PhysRevD.108.046014](#) - [arXiv:2112.10781](#)

2022 - **Chaos and pole-skipping in a simply spinning plasma**
[10.1007/JHEP02\(2023\)253](#) - [arXiv:2211.00016](#)

2020 - **Hydrodynamics of simply spinning black holes & hydrodynamics for spinning quantum fluids**
[10.1007/JHEP12\(2020\)112](#) - [arXiv:2007.04345](#)

2020 - **Resonating AdS soliton**
[10.1007/JHEP08\(2020\)136](#) - [arXiv:2006.12783](#)

2019 - **Dispersion relations in non-relativistic two-dimensional materials from quasinormal modes in Hōrava Gravity**
[10.1007/JHEP10\(2019\)087](#) - [arXiv:1905.11993](#)

2.2 Given Talks

2025 CDIO Asian Regional Meeting <i>Politeknik Negeri Batam</i>	August, 2025
Strings and Fields <i>Kyoto University</i>	July, 2025
Solitons at Work (INVITED) <i>Online</i>	November, 2024
Research Seminar <i>Tokyo Institute of Technology</i>	May, 2024
Research Seminar <i>Keio University</i>	May, 2024
Research Seminar <i>Chuo University</i>	May, 2024
Rising Researchers Seminar (INVITED) <i>Online</i>	April, 2024
International Workshop on Noncommutative Integrable Systems <i>Online</i>	March, 2024
Holotube (INVITED) <i>Online</i>	December, 2023
Online Workshop on Topological Solitons <i>Online</i>	September, 2023
Research Seminar <i>Tokyo University</i>	February, 2020
Research Seminar <i>Ochanomizu University</i>	February, 2020
Research Seminar <i>Chuo University</i>	February, 2020
Research Seminar <i>Würzburg University</i>	July, 2019
Frankfurt Institute for Advanced Studies <i>Würzburg University</i>	July, 2019
3rd Karl Schwarzschild Conference <i>Würzburg University</i>	July, 2017

3 Teaching Experience

Environmental Science - Spring 2025 - Present

National Institute of Technology Oyama College - *once per semester lecture on environmental science in English*

Special Lectures on Special Relativity - Spring 2024
 Yamagata University - *informal lectures taught in Japanese*

PH 255: Modern Physics Laboratory - Spring 2021
 The University of Alabama - *as a graduate teaching assistant*

PH 255: Modern Physics Laboratory - Fall 2020
 The University of Alabama - *as a graduate teaching assistant*

PH 102: General Physics II - Summer 2020
 The University of Alabama - *two classes were taught*

PH 125: Honors General Physics with Calculus I - Fall 2019
 The University of Alabama - *as a graduate teaching assistant*

PH 255: Modern Physics Laboratory - Fall 2018
 The University of Alabama - *as a graduate teaching assistant*

Tutoring Session (College Math and Physics) - Fall 2018 - Spring 2021
 Applied Tutoring - *taught privately for around 5hrs a week at the University of Alabama*

PH 125: Honors General Physics with Calculus I - Fall 2016
 The University of Alabama - *as a graduate teaching assistant*

PH 105: General Physics with Calculus I - Fall 2016 - Spring 2021
 The University of Alabama - *as a graduate teaching assistant for 0-2 classes per year*

PH 106: General Physics with Calculus II - Fall 2016 - Spring 2021
 The University of Alabama - *as a graduate teaching assistant for 0-2 classes per year*

4 Recognition and Honors

4.1 Awards

Japanese-Language Proficiency Test (JLPT) N2 Level Certification	2023
Japan Society for the Promotion of Science Fellowship <i>Yamagata University</i>	2023
Outstanding Research by a Master's Student <i>The University of Alabama</i>	2020
JASSO Scholarship <i>Tohoku University</i>	2014
Private Scholarship <i>Pagosa Springs High School</i>	2008

4.2 Memberships

Global Center Member <i>NIT Oyama College Global Center</i>	April 2025 - present
President <i>Physics and Astronomy Graduate Physics Association of the University of Alabama</i>	2017 - 2021
Secretary <i>Japanese American Cultural Exchange Club of the University of Alabama</i>	2017 - 2020